

Paper Practice 2.5 • WORKED KEY**Solving Equations Involving Absolute Value**

WORKED KEY (odd-numbered items) – Split $|\text{expression}| = k$ into $\text{expression} = k$ and $\text{expression} = -k$. If k is negative there is no solution. For a graph, the equation is $|x - \text{midpoint}| = \text{distance}$. Even-numbered items are yours to work.

Examples 1 and 2

Solve each equation. Then graph the solution set.

1. $|x + 3| = 2$

Answer: $x + 3 = 2$ or $x + 3 = -2 \rightarrow x = -1$ or $x = -5$; $\{-5, -1\}$. Graph: closed dots at -5 and -1 .

2. $|a - 4| = 3$

Solve each equation.

5. $|3v + 5| = 8$

Answer: $3v + 5 = 8 \rightarrow v = 1$; $3v + 5 = -8 \rightarrow 3v = -13 \rightarrow v = -\frac{13}{3}$. $\{1, -\frac{13}{3}\}$

6. $|t + 8| = 11$

7. $|4c + 2| = -2$

Answer: no solution – an absolute value is never negative.

8. $|2x + 11| = 3$

3. $|d - 4| = -2$

Answer: no solution – an absolute value is never negative. Graph: nothing to plot (empty set).

4. $|2w + 5| = 5$

9. $|5m + 1| = 9$

Answer: $5m + 1 = 9 \rightarrow m = \frac{8}{5}$; $5m + 1 = -9 \rightarrow m = -2$. $\{\frac{8}{5}, -2\}$

10. $|7p + 2| = -7$

11. $|r + 8| = 8$

Answer: $r + 8 = 8 \rightarrow r = 0$; $r + 8 = -8 \rightarrow r = -16$. $\{0, -16\}$

12. $|4w + 7| = 11$

Example 3

13. **PENCILS** A company manufactures pencils that are 7.5 inches long. To pass quality inspection, the length of a pencil must be within $\pm \frac{1}{8}$ inch of the desired length. Write and solve an absolute value equation to find the minimum and maximum acceptable lengths of a pencil.

Answer: $|x - 7.5| = \frac{1}{8} \rightarrow x - 7.5 = \pm 0.125 \rightarrow$ **minimum 7.375 in ($7\frac{3}{8}$ in), maximum 7.625 in ($7\frac{5}{8}$ in)**

14. **SURVEY** A group of students was asked about favorite colors, and 20% of the students said that their favorite color is red. There are about 3000 students in the school. If the margin of error is $\pm 2\%$, what is the range of the number of students who likely say that their favorite color is red?

Example 4

Write an equation involving absolute value for each graph.

15. dots at 0 and 4

Answer: midpoint 2, each dot 2 away $\rightarrow |x - 2| = 2$

16. dots at -3 and 3

17. dots at -5 and 1

Answer: midpoint -2 , each dot 3 away $\rightarrow |x + 2| = 3$

18. dots at -1 and 3

19. dots at -2 and 6

Answer: midpoint 2 , each dot 4 away $\rightarrow |x - 2|$
 $= 4$

20. dots at -7 and -3